$$\log\frac{\vee}{\Lambda} = \log\frac{\vee}{k^+} + \log\frac{k^+}{\Lambda} \quad \text{(shared, not counted twice)}$$

measured gluon

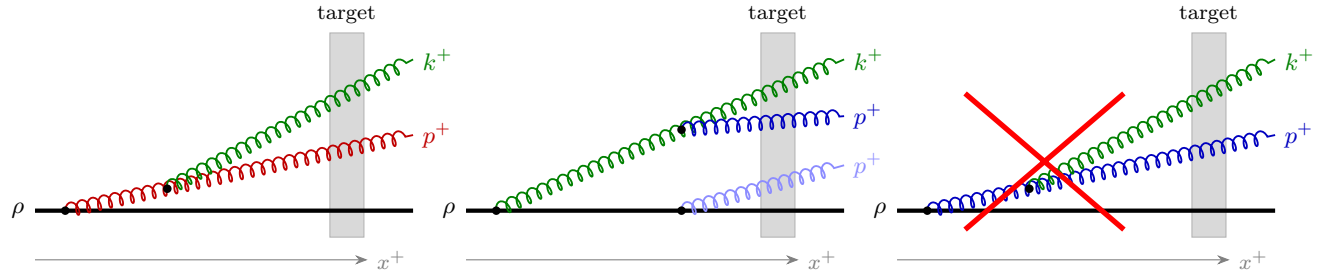
soft gluon **slower** than the measured one
$\Lambda < p^+ < k^+$: part of the **target** ⇒ **JIMWLK**

soft gluon **faster** than the measured one
$k^+ < p^+ < \vee$: part of the **projectile** ⇒ **BFKL**

$\log p^+$

Λ (target) $\quad k^+ \quad \vee$ (projectile)

$\log\frac{k^+}{\Lambda}$ (JIMWLK) $\qquad$ $\log\frac{\vee}{k^+}$ (BFKL)



(a) $k^+ < p^+ < \vee$: the soft gluon is formed first and *emits* the measured gluon
⇒ projectile evolution, **BFKL**, $\log\frac{\vee}{k^+}$

(b) $\Lambda < p^+ < k^+$: the soft gluon is formed last, next to the target, by ρ or by k^+
⇒ target evolution, **JIMWLK**, $\log\frac{k^+}{\Lambda}$

(c) $p^+ < k^+$: a softer gluon cannot emit the measured gluon (p^+ is conserved)
⇒ BFKL never gets $\log\frac{k^+}{\Lambda}$, so never $\log\frac{\vee}{\Lambda}$

$$\left.\frac{d^3N}{d^3k}\right|_{\text{LL}} = \underbrace{\log\frac{\vee}{\Lambda}\,\mathcal{J}}_{\text{coefficient of } \log\frac{\vee}{\Lambda}\text{: JIMWLK alone}} + \underbrace{\log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J})}_{\text{BFKL only with } \log\frac{\vee}{k^+}} = \log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}$$

$\mathcal{J}$ = JIMWLK ⊗ LO, $\quad \mathcal{B}$ = BFKL ⊗ LO; $\quad$ at mid rapidity, $k^+ = \sqrt{\vee\Lambda}$: $\ \frac{1}{2}\log\frac{\vee}{\Lambda}\,(\mathcal{J}+\mathcal{B})$